Naman Antony Menezes

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Education

Carnegie Mellon University, School of Computer Science 4.06/4.00
Masters of Science in Robotics Systems Development August 2023 - May 2025

R. V. College of Engineering 9.14/10.00
B.E. in Electronics and Communication Aug. 2018 - Aug. 2022

Experience

Software Engineer June 2025 - Present
NVIDIA Corporation California

- Trained MapGPT, a 600M-parameter VLM (DINOv2, Llama 3.1), using DDP on 64 H100s and GRPO RL
- Made model training 2x faster with an efficiency optimization, saving the company ~$500K annually
- Shipped MapGPT to production, mapping multiple US cities and labelling all of San Francisco in hours
- Owned a new human-in-the-loop map editor model end to end: concept, data, training and deployment at scale
- Built the vLLM production inference stack, scaling to thousands of GPU-hours on AWS Kubernetes
- Cut map generation cost ~5x by running inference only along road geometry instead of full tile grids
- Improved output precision and recall by 10% through inference-time scaling

Software Engineering Intern May 2024 - August 2024
NVIDIA Corporation [Docs] California

- Designed an exporter from NVIDIA's 3D occupancy maps to open 3D Tiles with geospatial alignment and LODs
- Scaled the architecture to run on massively parallel cloud servers, exporting city scale maps in under 5 hours
- Cut 3D Tile map size ~5x with meshoptimizer, enabling city-scale map visualization in a browser for the first time

Project Associate August 2022 - August 2023
Artificial Intelligence and Robotics Lab (AIRL), Indian Institute of Science [Demos] Bangalore

- Developed an Autonomous Driving Stack for GPS Denied Environments presented at CES 2024
- Responsible for developing the entire low level communication architecture over CAN Bus

Software Engineering Intern March 2022 - July 2022
Packet Forwarding Engine Team, Juniper Networks [Certificate] Bangalore

- Developed a Python Tool that drastically reduces time spent in debugging routers

Projects & Competitions

Steerable Collaborative Multi-Robot Policies, Actor Labs Act-athon [Video] September 2026

- Top 5 - fine-tuned Physical Intelligence's π0.7 VLA into one language-steerable policy for two robots
- Designed and built a butter-passing robot (mobile base, two lift arms) that collaborates with an SO-101

RoboHacks, Y Combinator [Video] April 2026

- Winner - Overall Prize and Scale AI Data Track ($2,500 and two Innate robots)
- Built cross-robot spatial memory turning iPhone video and narration into a shared 3D map for pick and place
- Integrated NVIDIA Parakeet ASR, 3D reconstruction, ICP, Gemma 3 VLM and an action chunking policy

Meta Reality Labs: Photorealistic Human Avatars using Drones [Website] [Video] Jan 2024

- Created a human following drone that avoids obstacles using DJI SDK and ROS2 on Mavic 3 Enterprise
- Trained a model on synthetically generated avatar models and demonstrated a real time full body codec avatar

GPS Denied Autonomous Navigation Algorithm [Demo] July 2023

- Coded a full stack autonomous navigation algorithm for GPS Denied Environments
- Developed LiDAR map to real world registration using particle filter leveraging PCL Library
- Implemented local and global planning algorithms resulting in real time obstacle avoidance and road alignment

2D LiDAR Only SLAM Algorithm July 2023

- Built real-time CUDA 2D LiDAR SLAM in C++/ROS1 with Fourier-Mellin frontend and SE-Sync backend

Publications and Patents

Single Drive Based Mapping and Scene Understanding Patent Application, NVIDIA 2026

- Co-inventor; assigned to NVIDIA; filed in the US

Multi Robot Path Planning Using Priority Based Algorithm Published IEEE CONECCT July 2022

- Created and open sourced an algorithm to plan for multiple robots with distinct start and goal points
- Simulated scenarios with multiple robots on Gazebo Square Park Map, Narrow Corridor, City Block Map

Technical Skills

Machine Learning: PyTorch, vLLM, VLMs, VLAs, DDP, GRPO, reinforcement learning, multi-node training, fine-tuning
Robotics: ROS1/ROS2, Gazebo, SLAM, sensor fusion, motion planning, PCL, LiDAR, point clouds
Languages and Infrastructure: Python, C/C++, CUDA, AWS, Kubernetes, Docker, Linux, Git, CMake